

Why should IQR is multiplied with 1.5 in percentile calculation?

It is a standard rule to identify potential outliers.

The multiplier 1.5 is used because it acts as a middle point to being too strict and too lenient. Where if multiplier 1.0

is used, it often flags too many data points as outliers and if multiplier 2.0 is used then it is too inclusive of data points.

The 1.5IQR rule defines the data range which covers 99.3% of data.

Identify Outliers

	Minimum	Q1	Median	Q3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

$IQR = Q3 - Q1$

Lesser range = $Q1 - 1.5 * IQR$

Greater range = $Q3 + 1.5 * IQR$

	Minimum	Q1	Median	Q3	Maximum	$IQR = Q3 - Q1$	Lesser range = $Q1 - 1.5 * IQR$	Greater range = $Q3 + 1.5 * IQR$
Day	32	56	74.5	82.5	99	26.5	16.25	122.25
Night	25.5	78	81	89	98	11	61.5	105.5

Summary for Day and Night classes:

Day:

As we have arrived with Lesser range as 16.25 and there is no value less than 16.25. Hence we are concluding that there is no Lesser Outlier in Day.

Also as we have Greater outlier range as 122.25 and there is no value greater than 122.25. Hence we are concluding that there is no Greater Outlier in Day.

Night:

As we have arrived with Lesser range as 61.5 and the minimum value of Night is less than 61.5, we conclude that the lesser outlier exists from 61.5 to 25.5.

Also as we have Greater outlier range as 105.5 and the maximum value is 98. Hence we are concluding that there is no Greater Outlier in Night.